LEVEL 2 · PART I

Networking

2

Goal: Build the networking foundation senior DevOps work depends on; explain exactly what happens from typing https://example.com to the application response.

RECOMMENDED TIME
2-4 weeks

TOPICS
11

PRACTICE / CAPSTONE
Trace a full HTTPS request end to end using dig, curl -v, ss and tcpdump, and write the explanation in your own words.

Topics

#	Topic	What you will learn
1	IP, MAC, ARP and DHCP **p.399**	addressing layers, how a host gets an address, ARP resolution
2	TCP, UDP and ICMP **p.411**	reliability, handshakes, when UDP, ping/traceroute via ICMP
3	IPv4, IPv6, CIDR and Subnets **p.423**	CIDR math, subnetting a VPC, IPv6 basics
4	Routing, NAT, Ports and Firewalls **p.435**	routing tables, default gateway, SNAT/DNAT, port ranges, stateful firewalls, iptables/nftables
5	DNS Resolution and Domain Architecture **p.449**	recursive vs authoritative, record types, TTL, zones, delegation
6	TCP Connections and TLS Handshakes **p.461**	3-way handshake, TLS 1.2/1.3 handshake, certificates, SNI
7	HTTP Methods and Status Codes **p.475**	GET/POST/PUT/PATCH/DELETE, idempotency, 2xx/3xx/4xx/5xx, headers
8	Reverse Proxies and Load Balancers **p.489**	Nginx/HAProxy/Envoy, health checks, balancing algorithms
9	L4 versus L7 Load Balancing **p.503**	transport vs application routing, NLB vs ALB, trade-offs

#	Topic	What you will learn
10	Network Debugging Tools **p.517**	ping traceroute curl dig nslookup ss tcpdump nc, a debugging decision tree
11	What Happens When You Visit https://example.com **p.531**	end-to-end synthesis of every previous topic
★	Level capstone project **p.549**	Trace a full HTTPS request end to end using dig, curl -v, ss and tcpdump, and write the explanation in your own words.

How to work through this level

1. Read each topic in order: *what it is* → *why it matters* → *real-world example* → *diagram* → *guide*.
2. Do **every task** in the topic's *Tasks* section and tick the box (`- [x]`) in your own fork or notes.
3. Answer the *Self-check questions* without looking, then reveal the answers.
4. Finish with the **capstone project**, which combines all topics of the level.
5. Move on only when you can do the capstone's *Prove it* task without the guide.

Topic flow

